

[BUG] Incorrect shot batching behaviour in ExpvalCost when optimize=True

<https://github.com/PennyLaneAI/pennylane/issues/1480>

ROW 105

[BUG] Incorrect shot batching behaviour in ExpvalCost when optimize=True #1480

New issue

Closed

#1897



trbromley opened on Jul 27, 2021

Expected behavior

Suppose we want to evaluate the expectation value of a Hamiltonian:

```
h = qml.Hamiltonian([1, 1], [qml.PauliZ(0), qml.PauliZ(1)])
```

We want to do this using batched-shots

```
shot_batch = [(100, 10)]
```

To do this in PennyLane, we use `ExpvalCost`. There are options based on the `optimize` flag, which sets whether to group commuting observables:

```
exp_unoptimized = qml.ExpvalCost(ansatz, h, dev, optimize=False)
exp_optimized = qml.ExpvalCost(ansatz, h, dev, optimize=True)
```

Evaluating the result with `optimize=False` gives the expected output shape:

```
>>> exp_unoptimized(args)
[1.92 1.9  1.98 1.92 1.96 1.92 1.92 1.94 1.86 1.9 ]
```

Actual behavior

When `optimize=False`, we get an unexpected output shape:

```
>>> exp_optimized(args)
[1.8 2. ]
```

Additional information

We may decide not to fix this given `ExpvalCost` may soon be replaced, and suggest users set `optimize=False`. On the other hand, `optimize=True` provides some impressive speedups and switching to `optimize=False` will likely prevent users being able to successfully optimize.

Related post: <https://discuss.pennylane.ai/t/differentiable-circuit-with-sampling/1190>

Source code

```
import pennylane as qml

shot_batch = [(100, 10)]
dev = qml.device("default.qubit", wires=2, shots=shot_batch)
h = qml.Hamiltonian([1, 1], [qml.PauliZ(0), qml.PauliZ(1)])

def ansatz(params, wires):
    qml.RX(params, wires=wires[0])

exp_unoptimized = qml.ExpvalCost(ansatz, h, dev, optimize=False)
exp_optimized = qml.ExpvalCost(ansatz, h, dev, optimize=True)

print(exp_unoptimized(0.4))
print(exp_optimized(0.4))
```

Tracebacks

No response

Assignees

antalszava

rmoyard

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

'ExpvalCost' when 'optimize=True' with shots batch
PennyLaneAI/pennylane

Notifications

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Participants



System information

Happens on dev branch of PL.



☒ I have searched existing GitHub issues to make sure the issue does not already exist.



trbromley added **bug** on Jul 27, 2021

github-actions assigned **antalszava** on Jul 27, 2021



trbromley on Jul 27, 2021

Contributor

Author



Do we need to edit the template to automatically format to Python? See below:

Source code

```
import pennylane as qml

shot_batch = [(100, 10)]
dev = qml.device("default.qubit", wires=2, shots=shot_batch)
h = qml.Hamiltonian([1, 1], [qml.PauliZ(0), qml.PauliZ(1)])

def ansatz(params, wires):
    qml.RX(params, wires=wires[0])

exp_unoptimized = qml.ExpvalCost(ansatz, h, dev, optimize=False)
exp_optimized = qml.ExpvalCost(ansatz, h, dev, optimize=True)

print(exp_unoptimized(0.4))
print(exp_optimized(0.4))
```



Otherwise, template was fun to use! The "System information" felt a bit unnecessary for this post, but I can understand why it's there as a required field.



mariaschuld on Jul 28, 2021

Contributor



@trbromley we are busy finding a better way to integrate `shot_batch`. Basically, it will always be ambivalent if we set it in the device, but use the device to do all sorts of things. The solution we are exploring is to do the grouping and shot distribution when creating the Hamiltonian (otherwise using a default).

So I personally suggest fixing this with the new VQE design!



rmoyard self-assigned this on Aug 20, 2021

rmoyard mentioned this on Nov 15, 2021

[ExpvalCost when optimize=True with shots batch #1897](#)



rmoyard on Nov 15, 2021

Contributor



[\[sc-8002\]](#)



ExpvalCost when optimize=True with shots batch #1897

[Code](#)

Merged rmoyard merged 7 commits into `master` from `expval_cost_shots` on Nov 16, 2021

Conversation 5 Commits 7 Checks 0 Files changed 3 +28 -9



rmoyard commented on Nov 15, 2021

Contributor

Context:

`ExpvalCost` returns wrong values when `optimize=True` because of the shape of results due to shots (e.g (8000,5)).

Description of the Change:

Change the function to take into account the shape of the results.

Benefits:

It now returns the correct values.

Related GitHub Issues:

Closes [#1480](#)



Reviewers

dime10



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[BUG] Incorrect shot batching behaviour in E...

Notifications

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2 participants



rmoyard added 2 commits 4 years ago

First version.

12fe0e2

Typo.

2e7e2c8



github-actions bot commented on Nov 15, 2021

Contributor

Hello. You may have forgotten to update the changelog!

Please edit [doc/releases/changelog-dev.md](#) with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.



rmoyard added 2 commits 4 years ago

Black

8c53c67

Changelog.

e642931

rmoyard requested a review from dime10 4 years ago